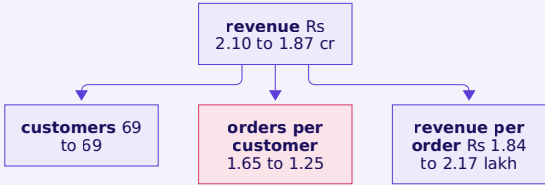


1 The tree, Q1 against Q2



The six chapters climb it in order: real, branch, segment, mix or rate, Marketing's claim, the memo.

Revenue fell 11.0 percent on closed quarters: 1.000 times 0.754 times 1.180 is 0.890. In behaviour the fall sits in Retail-Plus, where the same 22 members placed 26 orders against 51.

**Crux.** Confirm the drop on matched windows before you explain it.

2 Windows, like with like

| Window                         | Result            | Verdict                                        |
|--------------------------------|-------------------|------------------------------------------------|
| Q1 against Q2 cut at 15 Sep    | Down 25.9 percent | First and last dates show 13 weeks against 11. |
| Closed quarters, 13 weeks each | Down 11.0 percent | It is fair once Q2 has closed.                 |
| Same 11 weeks, Q2 still open   | Down 17.0 percent | Matched weeks while a quarter is open.         |
| Per week on the cut window     | Down 12.4 percent | A rate, said with its window.                  |
| Second route, by month         | Down 11.0 percent | The dates agree with the quarter field.        |

**Crux.** A rate without its denominator is a rumour.

3 The tree, as a bridge in rupees

| Step                                    | Rupees             |
|-----------------------------------------|--------------------|
| Q1 revenue                              | Rs 2,10,00,000     |
| Customers, 69 to 69                     | Rs 0               |
| Orders per customer, 28 fewer orders    | Minus Rs 51,57,895 |
| Revenue per order, Rs 33,231 more on 86 | Plus Rs 28,57,895  |
| Q2 revenue                              | Rs 1,87,00,000     |

**Crux.** Decompose along the tree: customers, orders per customer, revenue per order.

4 Missing is unknown

```
order.get("discount", 0) # a decision that looks like no decision
```

Read as zero, a blank counts as "no discount", so 50.0 percent of Q2's orders seem to carry one. Split the rest into recorded zeros and blanks, report the share where recorded, 71.7 percent, with the range the blanks allow, and bound the branch: Rs 150 on 86 orders is at most Rs 12,900.

**Crux.** Missing means unknown until someone chooses a default and writes down why.

5 Roll-up with weights

A function beats eight copied loops. A plain average of four segments gives 1.94 and 1.82; total orders over total customers gives 1.65 and 1.25, the figure a roll-up must reproduce.

**Crux.** Roll a rate up with its weights; never average the averages.

6 Mix against rate

Revenue per order rose Rs 33,231 and no segment rose 18 percent: about 69 percent is mix, since small Retail-Plus orders fell from 44.7 to 30.2 percent of orders.

**Crux.** A blended rate can move while no segment moves; split mix from rate.

7 Functions return

```
def tree_for(rows):
    ...
    return {"revenue": revenue, "orders": n, "customers": c}
```

A helper that prints for large changes returns **None**, and the filter drops it: four segments go in and two come back as **None**. Check with **None in changes.values()**. The id overlap answers Marketing's churn claim: 69 in both quarters, 0 lost, 0 new.

**Crux.** A function returns its answer; count the groups in and the groups out.

8 Hypothesis with evidence

| Test                | What the file shows                                                           |
|---------------------|-------------------------------------------------------------------------------|
| Timing              | Retail-Plus fell from July, before the late-August break the complaint dates. |
| Channel             | Web, store and app all fell, so the app did not fall first and alone.         |
| Ceiling             | At the pre-break pace the button explains at most about 4 orders.             |
| Evidence to ask for | The app's reorder logs by week, the release date, and the tier's change log.  |

**Crux.** Name the cause as a hypothesis, with its ceiling and the evidence that would settle it.

THE WORDS ON THIS SHEET

**Matched windows** Two periods of the same length compared, or both turned into a rate

**Rate** A numerator over a denominator in a stated window

**Like-for-like** A comparison that holds the outlets, the weeks and the weekends equal

**Decomposition** A change split along the tree into branches that multiply

**Bridge** A change in rupees moved one branch at a time from start to end

**Range** The largest value less the smallest, set by two orders alone

**Mix** Each segment's share of orders

**Rate part** The change inside segments, mix held still